

PONAALAGAR O K Software Centric Solution Provider

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SUMMARY

AI & Full-Stack Developer with experience building production-ready applications across eCommerce, Edge AI, and automation systems. Skilled in Python and system design, with proven impact through deployed products and hackathon-winning solutions.

TECHNICAL SKILLS

Languages — Python | Java | C

Databases — Vector DB | Firebase

Platforms — Linux | Git/GitHub | AWS | Hostinger

Tools & Frameworks — TensorFlow | Pytorch | LangChain | Anomalib | OpenCV

CS Fundamentals — Data Structures | Algorithms | OOP | DBMS | OS

EDUCATION

Sri Kumaran Matriculation Higher Secondary School, HSC
2022-2023 | Marks:85%

Kongu Engineering college, Bachelor's Degree in Artificial Intelligence and Machine Learning
2023-2027 | CGPA:8.38*

CERTIFICATIONS

Machine Learning Specialization
By Coursera

Ethical Hacking
By NPTEL

Affective computing
By NPTEL

Introduction to Large Language Models (LLMs)
By NPTEL

PROFESSIONAL EXPERIENCE

Freelancer, Outliers United
01/2025 – Present

Agentic AI Intern, Cube AI Solutions
07/2025 – 12/2025

Software Engineering Intern, LIGTHS INC
02/2026 – Present

ACHIEVEMENT

1st prize, SIH internal Hackathon

1st prize, Hacknovate Hackathon

Finalist, Smart India Hackathon

Best Innovation award, bytes india hackathon

MEMBER SHIPS IN ACADEMIC BODY

Institutions innovation council, Class coordinator
05/2024 – 05/2025

AI Coding Club, Executive member
05/2024 – 05/2025

TECHNICAL PROJECTS

Easivisi, 12/2025 - Present

- Built web platform for no-code image annotation and object detection model training via GUI.
- Implemented dataset management, training configs, inference modules, and TensorBoard monitoring in unified interface.
- **Tech Stack:** Python, Flask, YOLOv8, PyTorch

Fabspector, 10/2025 - Present

- Developed end-to-end Edge AI solution for textile quality control using NVIDIA Jetson Nano.
- Integrated real-time video processing with GPIO hardware actuators to mark fabric defects instantly.
- **Tech Stack:** Python, OpenCV, YOLOv8, NVIDIA TensorRT, Flask, GPIO.

Tekhel India, 09/2025 - 12/2025

- Built automated fitness assessment and re-counting system using MoveNet and BlazePose for real-time pose estimation.
- Integrated identity verification to prevent proxies and optimized video processing with TFLite quantization for edge deployment.
- **Tech Stack:** MoveNet, BlazePose, TFLite, Python

MedRAG, 01/2026 - Present

- Built a production-grade medical RAG system with multi-modal ingestion (PDF + image OCR via Groq Vision) and vector retrieval (Milvus, Qdrant, Pinecone, ChromaDB) for accurate, citation-backed outputs.
- Developed dual-mode querying (RAG + Retriever) with an interactive Flask dashboard for efficient clinical workflows.
- **Tech Stack:** Python, Flask, LangChain (LCEL), Groq LLMs, Milvus, Sentence-Transformers, OCR, PyPDF2